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Geometrical Optics

NEET Physics · Previous Year Questions with Answer Key

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MCQ (Single Correct Answer)	83 questions
Answer key	at the end

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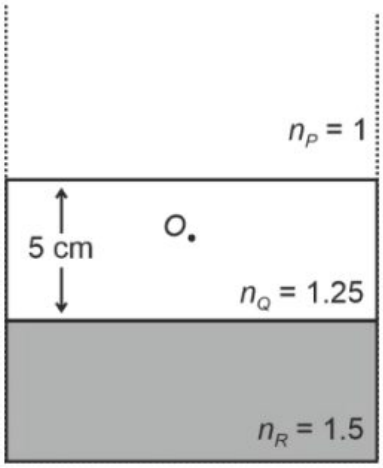
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MCQ (Single Correct Answer) 83 questions

Q1. RE-NEET 2026

Open solution

Consider three media P, Q and R with refractive indices $1, 1.25$, and 1.5 respectively. The medium Q having a thickness of 5 cm is placed between extended media P and R as shown in the figure. An object O is placed at the centre of medium Q. If viewed from medium P near the normal direction, the apparent depth of O is h_1. For similar observation from medium R, the apparent depth is h_2. The value of $|h_1 - h_2|$, in cm , is:

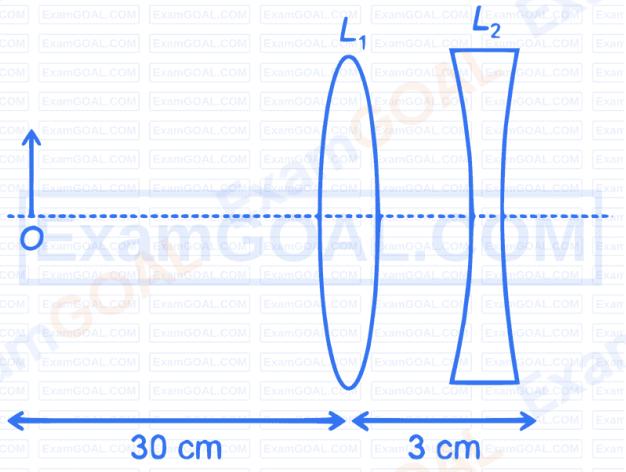


(A) 3 (B) 0 (C) 1 (D) 2

Q2. RE-NEET 2026

Open solution

The lens combination as shown in the figure, consists of two lenses, L_1 and L_2, of the focal lengths +10 cm and -10 cm , respectively. The position of the image formed is:



(A) 60 cm to the right of the concave lens (B) 20 cm to the left of the concave lens

(C) 60 cm to the left of the concave lens (D) 30 cm to the right of the concave lens

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Q3. RE-NEET 2026

Which of the following measurements require 'index correction'?

(A) Measurement of speed of sound using resonance tube

(B) Measurement of resistance of a wire using meter bridge

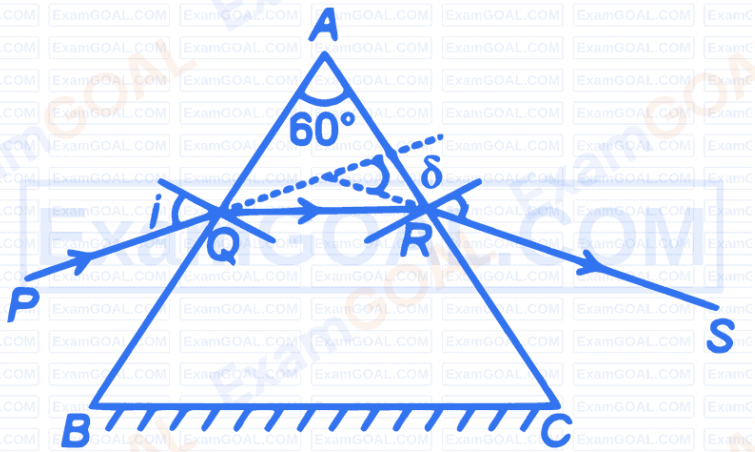
(C) Measurement of gravitational acceleration using simple pendulum

(D) Measurement of focal length of lenses using optical bench

Open solution

Q4. NEET 2026

A ray of monochromatic light is passing through an equilateral prism (ABC) as shown in the figure. The refracted ray (QR) is parallel to its base (BC) and the angle of incidence (i) is 50°. Then the angle of deviation (δ) is:



(A) 45° (B) 35° (C) 40° (D) 55°

Open solution

Q5. NEET 2026

In a concave lens, a ray of light emanating from the object parallel to the principal axis of the lens after refraction:

(A) passes through $2F$, which is the radius of curvature of the lens.

(B) appears to diverge from the first principal focus.

(C) emerges parallel to the principal axis.

(D) passes through the second principal focus.

Open solution

Q6. NEET 2025

A microscope has an objective of focal length 2 cm , eyepiece of focal length 4 cm and the tube length of 40 cm . If the distance of distinct vision of eye is 25 cm , the magnification in the microscope is

(A) 150 (B) 250 (C) 100 (D) 125

Open solution

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Q7. NEET 2025

In a certain camera, a combination of four similar thin convex lenses are arranged axially in contact. Then the power of the combination and the total magnification in comparison to the power (p) and magnification (m) for each lens will be, respectively

Open solution

(A) $4p$ and m^4 (B) p^4 and m^4 (C) $4p$ and $4m$ (D) p^4 and $4m$

Q8. NEET 2024 (Re-Examination)

Open solution

Given below are two statements:

Statement I : Image formation needs regular reflection and/or refraction.

Statement II : The variety in colour of objects we see around us is due to the constituent colours of the light incident on them.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Statement I is correct but statement II is incorrect

(B) Statement I is incorrect but Statement II is correct

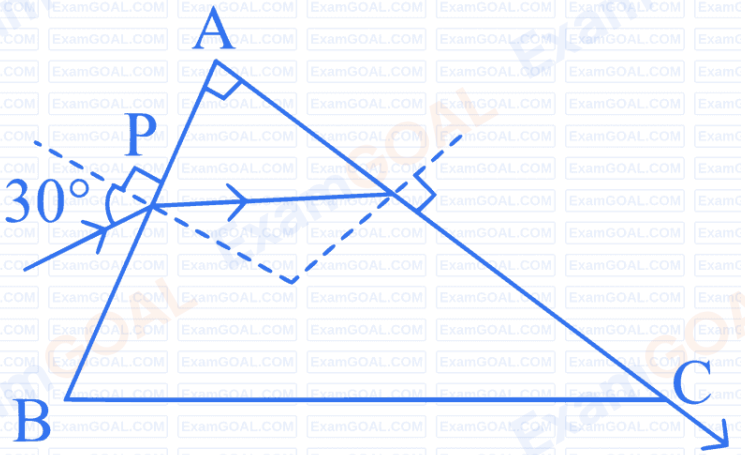
(C) Both Statement I and Statement II are correct

(D) Both Statement I and Statement II are incorrect

Q9. NEET 2024

Open solution

A light ray enters through a right angled prism at point P with the angle of incidence 30° as shown in figure. It travels through the prism parallel to its base BC and emerges along the face AC. The refractive index of the prism is:



(A) $\frac{\sqrt{5}}{4}$ (B) $\frac{\sqrt{5}}{2}$ (C) $\frac{\sqrt{3}}{4}$ (D) $\frac{\sqrt{3}}{2}$

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Q10. NEET 2024

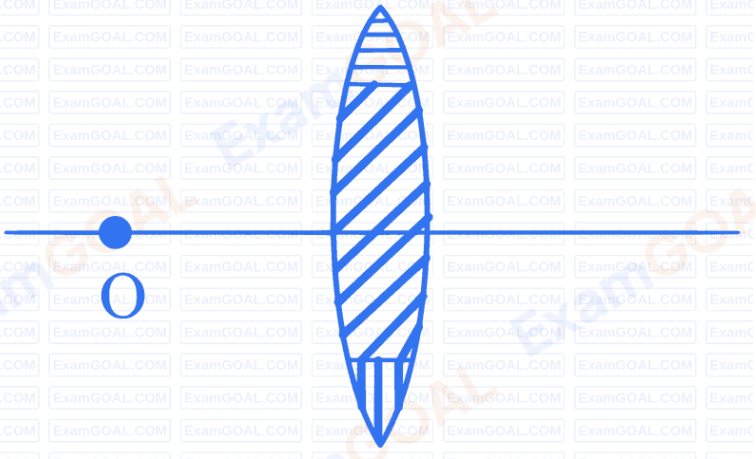
A small telescope has an objective of focal length $140\ \text{cm}$ and an eye piece of focal length $5.0\ \text{cm}$. The magnifying power of telescope for viewing a distant object is:

(A) 34 (B) 28 (C) 17 (D) 32

Open solution

Q11. NEET 2023 Manipur

A lens is made up of 3 different transparent media as shown in figure. A point object O is placed on its axis beyond $2f$. How many real images will be obtained on the other side?



(A) 2 (B) 1

(C) No image will be formed (D) 3

Open solution

Q12. NEET 2023 Manipur

ϵ_0 and μ_0 are the electric permittivity and magnetic permeability of free space respectively. If the corresponding quantities of a medium are $2\,\epsilon_0$ and $1.5\,\mu_0$ respectively, the refractive index of the medium will nearly be :

(A) $\sqrt{2}$ (B) $\sqrt{3}$ (C) 3 (D) 2

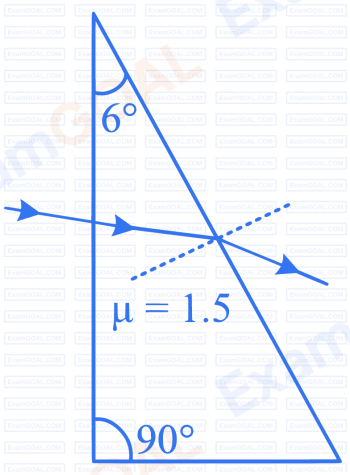
Open solution

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Q13. NEET 2023 Manipur

A horizontal ray of light is incident on the right angled prism with prism angle 6°. If the refractive index of the material of the prism is 1.5 , then the angle of emergence will be:

Open solution



(A) 9° (B) 10° (C) 4° (D) 6°

Q14. NEET 2023 Manipur

An object is mounted on a wall. Its image of equal size is to be obtained on a parallel wall with the help of a convex lens placed between these walls. The lens is kept at distance x in front of the second wall. The required focal length of the lens will be :

Open solution

(A) less than $\frac{x}{4}$ (B) more than $\frac{x}{4}$ but less than $\frac{x}{2}$

(C) $\frac{x}{2}$ (D) $\frac{x}{4}$

Q15. NEET 2023

Light travels a distance x in time t_1 in air and $10x$ in time t_2 in another denser medium. What is the critical angle for this medium?

Open solution

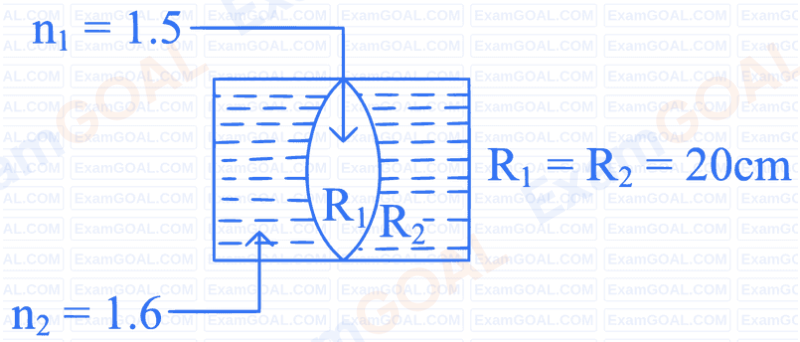
(A) $\sin^{-1}\left(\frac{10t_2}{t_1}\right)$ (B) $\sin^{-1}\left(\frac{t_1}{10t_2}\right)$

(C) $\sin^{-1}\left(\frac{10t_1}{t_2}\right)$ (D) $\sin^{-1}\left(\frac{t_2}{t_1}\right)$

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Q16. NEET 2023

In the figure shown here, what is the equivalent focal length of the combination of lenses (Assume that all layers are thin) ?



Open solution

(A) -40 cm (B) -100 cm (C) -50 cm (D) 40 cm

Q17. NEET 2023

Two thin lenses are of same focal lengths (f), but one is convex and the other one is concave. When they are placed in contact with each other, the equivalent focal length of the combination will be :

Open solution

(A) $f/4$ (B) $f/2$ (C) Infinite (D) Zero

Q18. NEET 2022 Phase 2

During a cloudy day, a primary and a secondary rainbow may be created, then the :

Open solution

(A) secondary rainbow is due to single internal reflection and is formed above the primary one.

(B) primary rainbow is due to double internal reflection and is formed above the secondary one.

(C) primary rainbow is due to double internal reflection and is formed below the secondary one.

(D) secondary rainbow is due to double internal reflection and is formed above the primary one.

Q19. NEET 2022 Phase 2

An astronomical refracting telescope is being used by an observer to observe planets in normal adjustment. The focal lengths of the objective and eye piece used in the construction of the telescope are 20 m and 2 cm respectively. Consider the following statements about the telescope :

Open solution

(a) The distance between the objective and eye piece is 20.02 m

(b) The magnification of the telescope is $(-)$ 1000

(c) The image of the planet is erect and diminished

(d) The aperture of eye piece is smaller than that of objective

The correct statements are :

(A) (a), (b) and (d) (B) (a), (b) and (c) (C) (b), (c) and (d) (D) (c), (d) and (a)

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Q20. NEET 2022 Phase 1

A biconvex lens has radii of curvature, 20 cm each. If the refractive index of the material of the lens is 1.5, the power of the lens is

(A) +2 D (B) +20 D (C) +5 D (D) Infinity

Open solution

Q21. NEET 2022 Phase 1

A light ray falls on a glass surface of refractive index $\sqrt{3}$, at an angle 60°. The angle between the refracted and reflected rays would be

(A) 30° (B) 60° (C) 90° (D) 120°

Open solution

Q22. NEET 2022 Phase 1

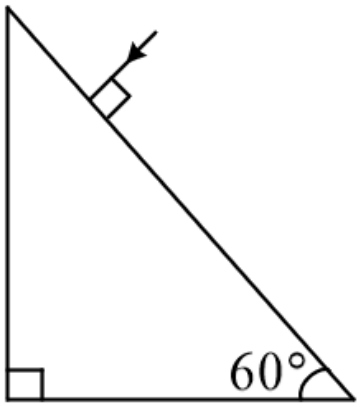
Two transparent media A and B are separated by a plane boundary. The speed of light in those media are 1.5×10^8 m/s and 2.0×10^8 m/s, respectively. The critical angle for a ray of light for these two media is

(A) $\sin^{-1}(0.500)$ (B) $\sin^{-1}(0.750)$ (C) $\tan^{-1}(0.500)$ (D) $\tan^{-1}(0.750)$

Open solution

Q23. NEET 2021

Find the value of the angle of emergence from the prism. Refractive index of the glass is $\sqrt{3}$



(A) 90° (B) 60° (C) 30° (D) 45°

Open solution

Q24. NEET 2021

A convex lens 'A' of focal length 20 cm and a concave lens 'B' of focal length 5 cm are kept along the same axis with a distance 'd' between them. If a parallel beam of light falling on 'A' leaves 'B' as a parallel beam, then the distance 'd' in cm will be :

(A) 30 (B) 25 (C) 15 (D) 50

Open solution

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Q25. NEET 2021

A lens of large focal length and large aperture is best suited as an objective of an astronomical telescope since :

Open solution

(A) a large aperture contributes to the quality and visibility of the images.

(B) a large area of the objective ensures better light gathering power.

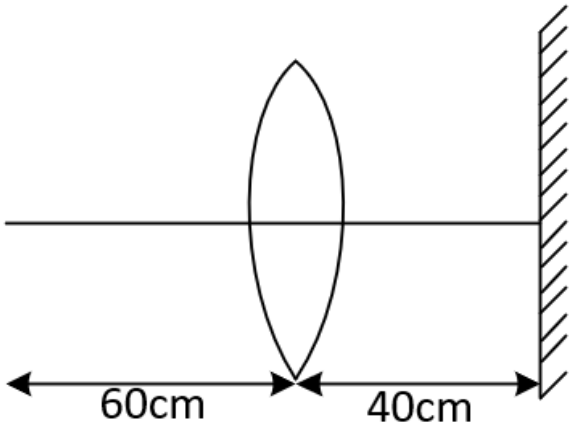
(C) a large aperture provides a better resolution.

(D) All the above.

Q26. NEET 2021

A point object is placed at a distance of 60 cm from a convex lens of focal length 30 cm. If a plane mirror were put perpendicular to the principal axis of the lens and at a distance of 40 cm from it, the final image would be formed at a distance of -

Open solution



(A) 20 cm from the plane mirror, it would be a virtual image.

(B) 20 cm from the lens, it would be a real image.

(C) 30 cm from the lens, it would be a real image.

(D) 30 cm from the plane mirror, it would be a virtual image.

Q27. NEET 2020 Phase 1

A ray is incident at an angle of incidence i on one surface of a small angle prism (with angle of prism A) and emerges normally from the opposite surface. If the refractive index of the material of the prism is μ, then the angle of incidence is nearly equal to :

Open solution

(A) $\frac{2A}{\mu}$ (B) μA (C) $\frac{\mu A}{2}$ (D) $\frac{A}{2\mu}$

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Q28. NEET 2020 Phase 1

The Brewster's angle i_b for an interface should be :

(A) $30^\circ < i_b < 45^\circ$

(B) $45^\circ < i_b < 90^\circ$

(C) $i_b = 90^\circ$

(D) $0^\circ < i_b < 30^\circ$

Open solution

Q29. NEET 2019

In total internal reflection when the angle of incidence is equal to the critical angle for the pair of media in contact, what will be angle of refraction?

(A) equal to angle of incidence

(B) 90°

(C) 180°

(D) 0°

Open solution

Q30. NEET 2019

Pick the **wrong** answer in the context with rainbow.

(A) An observer can see a rainbow when his front is towards the sun.

(B) Rainbow is a combined effect of dispersion refraction and reflection of sunlight.

(C) When the light rays undergo two internal reflections in a water drop, a secondary rainbow is formed.

(D) The order of colours is reversed in the secondary rainbow.

Open solution

Q31. NEET 2019

Two similar thin equi-convex lenses, of focal length f each, are kept coaxially in contact with each other such that the focal length of the combination is F_1. When the space between the two lenses is filled with glycerin (which has the same refractive index (μ = 1.5) as that of glass) then the equivalent focal length is F_2. The ratio $F_1 : F_2$ will be :

(A) 1 : 2

(B) 2 : 3

(C) 3 : 4

(D) 2 : 1

Open solution

Q32. NEET 2018

The refractive index of the material of a prism is $\sqrt{2}$ and the angle of the prism is 30°. One of the two refracting surfaces of the prism is made a mirror inwards, by silver coating. A beam of monochromatic light entering the prism from the other face will retrace its path (after reflection from the silvered surface) if its angle of incidence on the prism is

(A) 60^0

(B) 45^0

(C) 30^0

(D) zero

Open solution

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Q33. NEET 2018

An object is placed at a distance of 40 cm from a concave mirror of focal length 15 cm. If the object is displaced through a distance of 20 cm towards the mirror, the displacement of the image will be

(A) 30 cm away from the mirror
(B) 36 cm away from the mirror
(C) 30 cm towards the mirror
(D) 36 cm towards the mirror

Open solution

Q34. NEET 2018

An astronomical refracting telescope will have large angular magnification and high angular resolution, when it has an objective lens of

(A) small focal length and large diameter
(B) large focal length and small diameter
(C) large focal length and large diameter
(D) small focal length and small diameter

Open solution

Q35. NEET 2017

A beam of light from a source L is incident normally on a plane mirror fixed at a certain distance x from the source. The beam is reflected back as a spot on a scale placed just above the source L. When the mirror is rotated through a small angle θ, the spot of the light is found to move through a distance y on the scale. The angle θ is given by

(A) $\frac{y}{x}$
(B) $\frac{x}{2y}$
(C) $\frac{x}{y}$
(D) $\frac{y}{2x}$

Open solution

Q36. NEET 2017

A thin prism having refracting angle 10° is made of glass of refractive infex 1.42. This prism is combined with another thin prism of glass of refractive index 1.7. This combination producess dispersion without deviation. The refracting angle of second prism should be

(A) 6°
(B) 8°
(C) 10°
(D) 4°

Open solution

Q37. NEET 2016 Phase 2

Two identical glass (μ_g = 3/2) equiconvex lenses of focal length f each are kept in contact. The space between the two lenses is filled with water ($\mu_w = 4/3$). The focal length of the combination is

(A) $f/3$
(B) f
(C) $4f/3$
(D) $3f/4$

Open solution

Q38. NEET 2016 Phase 2

An air bubble in a glass slab with refractive index 1.5 (near normal incidence) is 5 cm deep when viewed from one surface and 3 cm deep when viewed from the opposite face. The thickness (in cm) of the slab is

(A) 8
(B) 10
(C) 12
(D) 16

Open solution

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Q39. NEET 2016 Phase 2

A person can see clearly objects only when they lie between 50 cm and 400 cm from his eyes . In order to increase the maximum distance of distinct vision to infinity, the type and power of the correcting lens, the person has to use, will be

Open solution

(A) convex, + 2.25 diopter
(B) concave, −0.25 diopter
(C) concave, −0.2 diopter
(D) convex, + 0.15 diopter

Q40. NEET 2016 Phase 1

Match the corresponding entries of column 1 with column 2. [Where m is the magnification produced by the mirror]

Open solution

Column 1			Column 2	
(A)	m = - 2		(p)	Convex mirror
(B)	m = $-\frac{1}{2}$		(q)	Concave mirror
(C)	m = +2		(r)	Real image
(D)	m = $+\frac{1}{2}$		(s)	Virtual image

(A) A $\rightarrow$ p and s; B $\rightarrow$ q and r; C $\rightarrow$ q and s; D $\rightarrow$ q and r
(B) A $\rightarrow$ r and s; B $\rightarrow$ q and s; C $\rightarrow$ q and r; D $\rightarrow$ p and s
(C) A $\rightarrow$ q and r; B $\rightarrow$ q and r; C $\rightarrow$ q and s; D $\rightarrow$ p and s
(D) A $\rightarrow$ p and r; B $\rightarrow$ p and s; C $\rightarrow$ p and q; D $\rightarrow$ r and s

Q41. NEET 2016 Phase 1

A astronomical telescope has objective and eyepiece of focal lengths 40 cm and 4 cm respectively. To view an object 200 cm away from the objective, the lenses must be separated by a distance

Open solution

(A) 50.0 cm (B) 54.0 cm (C) 37.3 cm (D) 46.0 cm

Q42. NEET 2016 Phase 1

The angle of incidence for a ray of light at a refracting surface of a prism is 45°. The angle of prism is 60°. If the ray suffers minimum deviation through the prism, the angle of minimum deviation and refractive index of the material of the prism respectively, are

Open solution

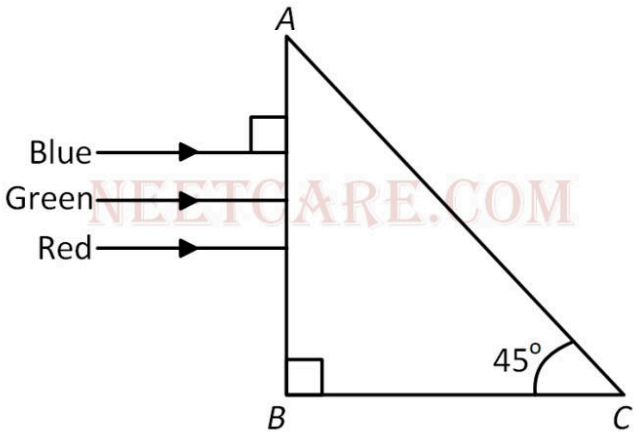
(A) $45^o; \sqrt{2}$ (B) $30^o; \frac{1}{\sqrt{2}}$ (C) $45^o; \frac{1}{\sqrt{2}}$ (D) $30^o; \sqrt{2}$

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Q43. AIPMT 2015

A beam of light consisting of red, green and blue colours is incident on a right angled prism. The refractive index of the material of the prism for the above red, green and blue wavelengths are 1.39, 1.44 and 1.47 respectively.

Open solution



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The prism will

(A) not separate the three colours at all

(B) separate the red colour part from the green and blue colours

(C) separate the blue colour part from the red and green colours

(D) separate all the three colours from one another

Q44. AIPMT 2015

In an astronomical telescope in normal adjustment a straight black line of length L is drawn on inside part of objective lens. The eye-piece forms a real image of this line. The length of this image is I. The magnification of the telescope is

Open solution

(A) $\frac{L+I}{L-I}$ (B) $\frac{L}{I}$ (C) $\frac{L}{I}+1$ (D) $\frac{L}{I}-1$

Q45. AIPMT 2015 Cancelled Paper

Two identical thin plano-convex glass lenses (refractive index 1.5) each having radius of curvature of 20 cm are placed with their convex surfaces in contact at the centre. The intervening space is filled with oil of refractive index 1.7. The focal length of the combination is

Open solution

(A) $-$ 50 cm (B) 50 cm (C) $-$ 20 cm (D) $-$ 25 cm

Q46. AIPMT 2015 Cancelled Paper

The refracting angle of a prism is A, and refractive index of the material of the prism is cot (A/2). The angle of minimum deviation is

Open solution

(A) $90^o - A$ (B) $180^o + 2A$ (C) $180^o - 3A$ (D) $180^o - 2A$

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Q47. AIPMT 2014

If the focal length of objective lens is increased then magnifying power of

(A) microscope will increase but that of telescope decrease.

(B) microscope and telescope both will increase.

(C) microscope and telescope both will decrease.

(D) microscope will decrease but that of telescope will increase.

Open solution

Q48. AIPMT 2014

The angle of a prism is A. One of its refracting surfaces is silvered. Light rays falling at an angle of incidence 2 A on the first surface returns back through the same path after suffering reflection at the silvered surface.

The refractive index μ, of the prism is

(A) 2sinA (B) 2cosA (C) $\frac{1}{2}$cosA (D) tanA

Open solution

Q49. NEET 2013 (Karnataka)

Two plane mirrors are inclined at 70°. A ray incident on one mirror at angle, θ after reflection falls on second mirror and is reflected from there parallel to first mirror. The value of θ is

(A) 45° (B) 30° (C) 55° (D) 50°

Open solution

Q50. NEET 2013

A plano convex lens fits exactly into a plano concave lens. Their plane surfaces are parallel to each other. If lenses are made of different materials of refractive indices μ_1 and μ_2 and R is the radius of curvature of the curved surface of the lenses, then the focal length of the combination is

(A) $\frac{R}{(\mu_1-\mu_2)}$ (B) $\frac{2R}{(\mu_2-\mu_1)}$

(C) $\frac{R}{2(\mu_1+\mu_2)}$ (D) $\frac{R}{2(\mu_1-\mu_2)}$

Open solution

Q51. NEET 2013

For a normal eye, the cornea of eye provides a converging power of 40 D and the least converging power of the eye lens behind the cornea is 20 D. Using this information, the distance between the retina and the cornea-eye lens can be estimated to be

(A) 1.67 cm (B) 1.5 cm (C) 5 cm (D) 2.5 cm

Open solution

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Q52. AIPMT 2012 Mains

For the angle of minimum deviation of a prism to be equal to its refracting angle, the prism must be made of a material whose refractive index.

(A) lies between $\sqrt{2}$ and 1
(B) lies between 2 and $\sqrt{2}$
(C) is less than 1
(D) is greater than 2

Open solution

Q53. AIPMT 2012 Mains

A rod of length 10 cm lies along the principal axis of a concave mirror of focal length 10 cm in such a way that its end closer to the pole is 20 cm away from the mirror. The length of the image is

(A) 10 cm
(B) 15 cm
(C) 2.5 cm
(D) 5 cm

Open solution

Q54. AIPMT 2012 Prelims

When a biconvex lens of glass having refractive index 1.47 is dipped in a liquid, it acts as a plane sheet of glass. This implies that the liquid must have refractive index

(A) equal to that of glass
(B) less than one
(C) greater than that of glass
(D) less than that of glass

Open solution

Q55. AIPMT 2012 Prelims

A ray of light is incident at an angle of incidence i, on one face of a prism of angle A (assumed to be small) and emerges normally from the opposite face. If the refractive index of the prism is μ. the angle of incidence i, is nearly equal to

(A) μA
(B) $\frac{\mu A}{2}$
(C) $\frac{A}{\mu}$
(D) $\frac{A}{2\mu}$

Open solution

Q56. AIPMT 2012 Prelims

A concave mirror of focal length f_1 is placed at a distance of d from a convex lens of focal length f_2. A beam of light coming from infinity and falling on this convex lens – concave mirror combination returns to infinity. The distance d must equal

(A) $f_1 + f_2$
(B) $-f_1 + f_2$
(C) $2f_1 + f_2$
(D) $-2f_1 + f_2$

Open solution

Q57. AIPMT 2012 Prelims

The magnifying power of a telescope is 9. When it is adjusted for parallel rays the distance between the objective and eyepiece is 20 cm. The focal length of lenses are

(A) 10 cm, 10 cm
(B) 15 cm, 5 cm
(C) 18 cm, 2 cm
(D) 11 cm, 9 cm

Open solution

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Q58. AIPMT 2011 Mains

A thin prism of angle 15° made of glass of refractive index μ_1 = 1.5 is combined with another prism of glass of refractive index μ_2 = 1.75. The combination of the prisms produces dispersion without deviation. The angle of the second prism should be

(A) 5° (B) 7° (C) 10° (D) 12°

Open solution

Q59. AIPMT 2011 Mains

A converging beam of rays is incident on a diverging lens. Having passed through the lens the rays intersect at a point 15 cm from the lens on the opposite side. If the lens is removed the point where the rays meet will move 5 cm closer to the lens. The focal length of the lens is

(A) 5 cm (B) − 10 cm (C) 20 cm (D) − 30 cm

Open solution

Q60. AIPMT 2011 Prelims

Which of the following is not due to total internal reflection ?

(A) Working of optical fibre

(B) difference between apparent and real depth of a pond

(C) Mirage on hot summer days

(D) Brilliance of diamond

Open solution

Q61. AIPMT 2011 Prelims

A biconvex lens has a radius of curvature of magnitude 20 cm. Which one of the following options best describes the image formed of an object of height 2 cm placed 30 cm from the lens ?

(A) Virtual, upright, height = 1 cm (B) Virtual, upright, height = 0.5 cm

(C) Real, inverted, height = 4 cm (D) Real, inverted, height = 1 cm

Open solution

Q62. AIPMT 2010 Mains

The speed of light in media M_1 and M_2 are 1.5×10^8 m/s and 2.0×10^8 m/s respectively. A ray of light enters from medium M_1 to M_2 at an incidence angle i. If the ray suffers total internal reflection, the value of i is

(A) Equal to $\sin^{-1}\left(\frac{2}{3}\right)$

(B) Equal to or less than $\sin^{-1}\left(\frac{3}{5}\right)$

(C) Equal to or greater than $\sin^{-1}\left(\frac{3}{4}\right)$

(D) Less than $\sin^{-1}\left(\frac{2}{3}\right)$

Open solution

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Q63. AIPMT 2010 Mains

A ray of light is incident on a 60° prism at the minimum deviation position. The angle of refraction at the first face (i.e., incident face) of the prism is

(A) zero (B) 30° (C) 45° (D) 60°

Open solution

Q64. AIPMT 2010 Prelims

A ray of light travelling in a transparent medium of refractive index μ, falls on a surface separating the medium from air at an angle of incidence of 45°. For which of the following value of μ the ray can undergo total internal reflection ?

(A) μ = 1.33 (B) μ = 1.40 (C) μ = 1.50 (D) μ = 1.25

Open solution

Q65. AIPMT 2010 Prelims

A lens having focal length f and aperture of diameter d forms an image of intensity I. Aperture of diameter $\frac{d}{2}$ in central region of lens is covered by a black paper. Focal length of lens and intensity of image now will be respectively

(A) f and $\frac{I}{4}$ (B) $\frac{3f}{4}$ and $\frac{I}{2}$
(C) f and $\frac{3I}{4}$ (D) $\frac{f}{2}$ and $\frac{I}{2}$

Open solution

Q66. AIPMT 2008

Two thin lenses of focal lengths f_1 and f_2 are in contact and coaxial. The power of the combination is

(A) $\frac{f_1+f_2}{2}$ (B) $\frac{f_1+f_2}{f_1 f_2}$
(C) $\sqrt{\frac{f_1}{f_2}}$ (D) $\sqrt{\frac{f_2}{f_1}}$

Open solution

Q67. AIPMT 2008

A boy is trying to start a fire by focusing sunlight on a piece of paper using an equiconvex lens of focal length 10 cm. The diameter of the sun is 1.39×10^9 m and its mean distance from the earth is 1.5×10^{11} m. What is the diameter of the sun's image on the paper ?

(A) 6.5×10^{-5} m (B) 12.4×10^{-4} m (C) 9.2×10^{-4} m (D) 6.5×10^{-4} m

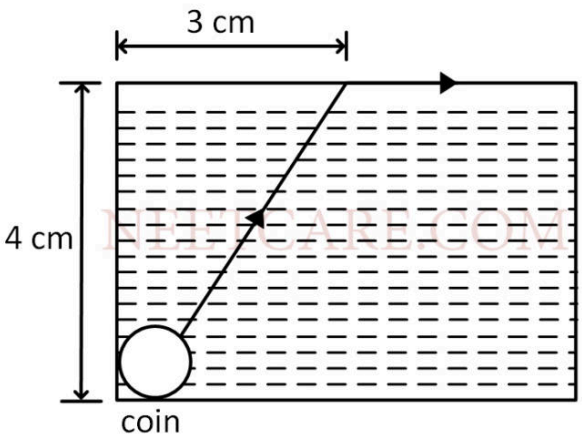
Open solution

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Q68. AIPMT 2007

A small coin is resting on the bottom of a beaker filled with liquid . A ray of light from the coin travels upto the surface of the liquid and moves along its surface. How fast is the light travelling in the liquid ?

Open solution



(A) 2.4×10^8 m/s (B) 3.0×10^8 m/s (C) 1.2×10^8 m/s (D) 1.8×10^8 m/s

Q69. AIPMT 2006

A microscope is focussed on a mark on a piece of paper and then a slab of glass of thickness 3 cm and refractive index 1.5 is placed over the mark. How should the microscope be moved to get the mark in focus again ?

Open solution

(A) 2 cm upward (B) 1 cm upward (C) 4.5 cm downward (D) 1 cm downward

Q70. AIPMT 2006

A convex lens and a concave lens, each having same focal length of 25 cm, are put in contact to form a combination of lenses. The power in diopters of the combination is

Open solution

(A) zero (B) 25 (C) 50 (D) infinite.

Q71. AIPMT 2004

The refractive index of the material of a prism is $\sqrt{2}$ and its refracting angle is 30°. One of the refracting surfaces of the prism is made a mirror inwards. A beam of monochromatic light entering the prism from the other face with retrace its path after reflection from the mirrored surface if its angle of incidence on the prism is

Open solution

(A) 45° (B) 60° (C) 0 (D) 30°

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Q72. AIPMT 2004

A beam of light composed of red and green ray is incident obliquely at a point on the face of rectangular glass slab. When coming out on the opposite parallel face, the red green ray emerge from

Open solution

(A) Two points propagating in two different non parallel directions

(B) Two points propagating in two different parallel directions.

(C) One point propagating in two different directions.

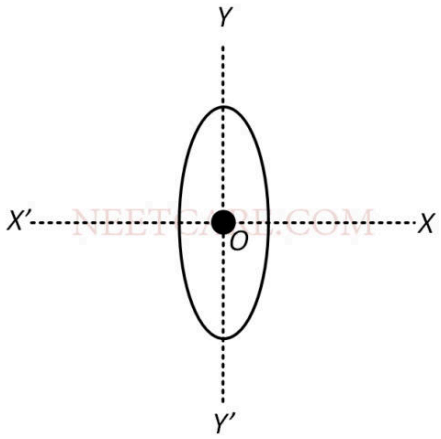
(D) One point propagating in the same firections.

Q73. AIPMT 2003

An equiconvex lens is cut into two halves along (i) XOX' and (ii) YOY' as shown in the figure. Let f, f', f'' be the focal lengths of the complete lens, of each half in case (i), and of each half in case (ii), respectively.

Open solution

Choose the correct statement from the following



(A) $f' = f, f'' = 2f$ (B) $f' = 2f, f'' = f$ (C) $f' = f, f'' = f$ (D) $f' = 2f, f'' = 2f$

Q74. AIPMT 2003

A convex lens is dipped in a liquid whose refractive index is equal to the refractive index of the lens. Then its focal length will

Open solution

(A) become zero (B) become infinite

(C) become small, but non-zero (D) remain unchanged

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Q75. AIPMT 2002

A bulb is located on a wall. Its image is to be obtained on a parallel wall with the help of convex lens. The lens is placed at a distance d ahead of second wall, then required focal length will be

Open solution

(A) only $\frac{d}{4}$
(B) only $\frac{d}{2}$
(C) more than $\frac{d}{4}$ but less than $\frac{d}{2}$
(D) less than $\frac{d}{4}$

Q76. AIPMT 2002

Diameter of human eye lens is 2 mm. What will be the minimum distance between two points to resolve them, which are situated at a distance of 50 meter from eye. The wavelength of light is 5000 $\overset{\circ}{A}$

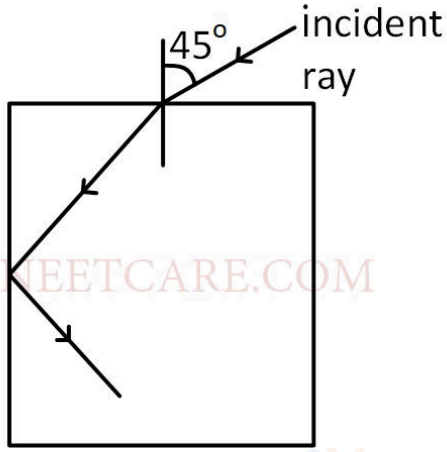
Open solution

(A) 2.32 m
(B) 4.28 mm
(C) 1.25 cm
(D) 12.48 cm

Q77. AIPMT 2002

For the given incident ray as shown in figure, the condition of total internal reflection of this ray the required refractive index of prism will be

Open solution



(A) $\frac{\sqrt{3}+1}{2}$
(B) $\frac{\sqrt{2}+1}{2}$
(C) $\sqrt{\frac{3}{2}}$
(D) $\sqrt{\frac{7}{6}}$

Q78. AIPMT 2001

Optical fibre are based on

Open solution

(A) total internal reflection
(B) less scattering
(C) refraction
(D) less absorption coefficient.

Q79. AIPMT 2001

A disc is placed on a surface of pond which has refractive index 5/3. A source of light is placed 4 m below the surface of liquid. The minimum radius of disc needed so that light is not coming out is,

Open solution

(A) ∞
(B) 3 m
(C) 6 m
(D) 4 m

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Q80. AIPMT 2000

A bubble in glass slab (μ = 1.5) when viewed from one side appears at 5 cm and 2 cm from other side, then thickness of slab is

(A) 3.75 cm (B) 3 cm (C) 10.5 cm (D) 2.5 cm.

Open solution

Q81. AIPMT 2000

A tall man of height 6 feet, want to see his full image. Then required minimum length of the mirror will be

(A) 12 feet (B) 3 feet (C) 6 feet (D) any length

Open solution

Q82. AIPMT 2000

For a plano convex lens (μ = 1.5) has radius of curvature 10 cm. It is silvered on its plane surface. Find focal length after silvering

(A) 10 cm (B) 20 cm (C) 15 cm (D) 25 cm

Open solution

Q83. AIPMT 2000

Rainbow is formed due to

(A) scattering and refraction (B) internal reflection and dispersion
(C) reflection only (D) diffraction and dispersion.

Open solution

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Answer Key

1. **C**	2. **C**	3. **D**	4. **C**	5. **B**
6. **D**	7. **A**	8. **C**	9. **B**	10. **B**
11. **D**	12. **B**	13. **A**	14. **C**	15. **C**
16. **B**	17. **C**	18. **D**	19. **A**	20. **C**
21. **C**	22. **B**	23. **B**	24. **C**	25. **D**
26. **A**	27. **B**	28. **B**	29. **B**	30. **A**
31. **A**	32. **B**	33. **B**	34. **C**	35. **D**
36. **A**	37. **D**	38. **C**	39. **B**	40. **C**
41. **B**	42. **D**	43. **B**	44. **B**	45. **A**
46. **D**	47. **D**	48. **B**	49. **D**	50. **A**
51. **A**	52. **B**	53. **D**	54. **A**	55. **A**
56. **C**	57. **C**	58. **C**	59. **D**	60. **B**
61. **C**	62. **C**	63. **B**	64. **C**	65. **C**
66. **B**	67. **C**	68. **D**	69. **B**	70. **A**
71. **A**	72. **B**	73. **A**	74. **B**	75. **B**
76. **C**	77. **C**	78. **A**	79. **B**	80. **C**
81. **B**	82. **A**	83. **B**		

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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